
Environmental Non-Screening and Finite-Size Stabilization in Z_6 QCD Chronometry

Nonlinear Earth, Sun and laboratory profiles for the protected ratio mode

Main conclusion. For the v0.6 benchmark, exact nonlinear spherical solutions for Earth, the Sun and a 5 cm tungsten source remain on the unscreened thick-shell branch. The largest nonlinear correction is 1.41×10^{-18} . Although the solar core lies above the spinodal density of the corresponding infinite homogeneous medium, the strongest solar sub-region reaches only

$$\frac{q_{\max}}{q_{\text{conv}}} = 1.02 \times 10^{-12}.$$

Finite-size gradient energy prevents phase conversion. The visible 2/27 QCD response therefore survives without environmental charge suppression; the benchmark signal is small because the coupling is small, not because matter screens it.

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Benchmark:	$M = 10$ TeV, $f_a = M_P$, $\epsilon = 10^{-6}$, $m_a = 3.80 \times 10^{-21}$ eV
Methods:	Exact nonlinear spherical boundary-value solutions, local-spinodal analysis, exterior-capacity bound and numerical verification
Principal result:	No thin shell and no matter-driven adjacent-phase core for Earth, Sun or laboratory sources

Abstract

We solve the nonlinear finite-density field equation of the Z_6 -protected QCD chronometry model for Earth, solar and laboratory sources. Visible matter contributes through the threshold law $m_A(x)/m_A(\pi/2) = [1 - \epsilon \cos x]^{p_A}$, with $p_A \leq 2/27$. The local homogeneous potential has a spinodal at $t_* = 0.1727234$, corresponding to 46.25 g cm^{-3} for the benchmark. The solar core exceeds this density, so an infinite medium would lose the metastable branch connected to $x = \pi/2$. A finite body, however, must also pay gradient energy. We derive a necessary spherical phase-conversion condition $q(R_c) > q_{\text{conv}} = 0.633135$, where $q(R_c) = M(< R_c)p\epsilon/(4\pi f_a^2 R_c)$. The largest value in the modeled Sun is 6.43×10^{-13} . Exact adaptive boundary-value solutions give screening factors equal to unity to numerical precision for all three sources. The protected $2/27$ QCD response is therefore not screened, while source-induced clock shifts remain small because $d_g = 7.41 \times 10^{-8}$. We identify cosmological vacuum/domain selection, rather than terrestrial or solar screening, as the next decisive problem.

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1 Main conclusion

For the v0.6 benchmark

$$M = 10 \text{ TeV}, \quad f_a = M_{\text{P}}, \quad \epsilon = 10^{-6},$$

with

$$m_a = 3.79715 \times 10^{-21} \text{ eV}, \quad \lambda_a = m_a^{-1} = 347.38 \text{ AU},$$

the exact nonlinear spherical field equation was solved for an Earth density model, a centrally concentrated solar model, and a uniform 5 cm tungsten sphere. All three bodies are on the **unscreened thick-shell branch**:

$$\mathcal{F}_{\text{screen}} = 1$$

to the numerical accuracy of the solve. The largest bounded nonlinear correction is 1.41×10^{-18} for the Sun.

The solar core is locally dense enough that an infinite homogeneous medium would remove the metastable $x = \pi/2$ minimum. Nevertheless, the Sun cannot move the field into the adjacent phase. Its strongest concentric sub-region has scalar compactness only

$$\frac{q_{\text{max}}}{q_{\text{conv}}} = 1.02 \times 10^{-12}.$$

The finite-size gradient cost overwhelms the local matter preference. Thus the finite-density effect does **not** screen the visible $2/27$ QCD threshold response into oblivion. It also does not amplify it: the benchmark remains weak because its unscreened coupling is small.

2 Model and environmental coupling

Let

$$x \equiv \frac{a}{f_a}$$

be the compact ratio mode. For the protected Z_6 construction, take the vacuum potential

$$V_{Z_6}(x) = V_0 (1 + \cos 6x), \quad V_0 = \frac{m_a^2 f_a^2}{36}.$$

Choose the asymptotic vacuum

$$x_\infty = \frac{\pi}{2}.$$

The visible vectorlike threshold gives, at leading one-loop order,

$$\frac{\Lambda_3(x)}{\Lambda_3(x_\infty)} = (1 - \epsilon \cos x)^{2/27}.$$

For a macroscopic body A , write the scalar-dependent mass as

$$A_A(x) \equiv \frac{m_A(x)}{m_A(x_\infty)} = (1 - \epsilon \cos x)^{p_A},$$

where

$$p_A = \frac{2}{27} Q_{\Lambda,A}.$$

Here $Q_{\Lambda,A}$ is the fraction of the body's logarithmic mass response carried by the low-energy QCD scale. The calculations use

$$p \equiv \frac{2}{27},$$

which is a conservative upper-envelope approximation $Q_{\Lambda,A} = 1$. Nuclear and electromagnetic corrections generally reduce the effective macroscopic charge and therefore reduce screening relative to this benchmark.

For conserved nonrelativistic rest density $\rho_{A*}(\mathbf{x})$, the effective potential is

$$V_{\text{eff}}(x, \mathbf{x}) = V_0(1 + \cos 6x) + \sum_A \rho_{A*}(\mathbf{x}) A_A(x).$$

The static equation is

$$\nabla^2 x = -\frac{m_a^2}{6} \sin 6x + \sum_A \frac{\rho_{A*}}{f_a^2} p_A \epsilon \sin x (1 - \epsilon \cos x)^{p_A-1}.$$

Boundary conditions for an isolated spherical source are

$$x'(0) = 0, \quad x(r \rightarrow \infty) = x_\infty.$$

This equation is nonlinear in both the periodic vacuum force and the matter coupling. No thin-shell or linear-response assumption is used in the numerical solution.

3 Dimensionless spherical problem

For a source of mass M , radius R , and density profile $\rho(r)$, define

$$s = \frac{r}{R}, \quad \mu = m_a R,$$

and the normalized density shape

$$D(s) = \frac{4\pi R^3 \rho(Rs)}{M}, \quad \int_0^1 D(s) s^2 ds = 1.$$

The source strength is

$$q_0 = \frac{Mp\epsilon}{4\pi f_a^2 R}.$$

Using $G = (8\pi M_P^2)^{-1}$, this can be written as

$$q_0 = 2p\epsilon \left(\frac{M_P}{f_a} \right)^2 \Phi_N, \quad \Phi_N = \frac{GM}{Rc^2}.$$

Let

$$x = x_\infty + q_0 u(s).$$

Inside the source, the exact scaled equation becomes

$$u'' + \frac{2}{s}u' = \mu^2 u \operatorname{sinc}(6q_0 u) + D(s) \cos(q_0 u) [1 + \epsilon \sin(q_0 u)]^{p-1}.$$

Outside, the field approaches the Yukawa form

$$x - x_\infty = -q_{\text{eff}} \frac{R}{r} e^{-m_a(r-R)}.$$

At the surface this gives the exact Robin condition

$$u'(1) + (1 + \mu)u(1) = 0.$$

The linearized problem is

$$u'' + \frac{2}{s}u' = \mu^2 u + D(s).$$

The screening factor used below is

$$\mathcal{F}_{\text{screen}} = \frac{u_{\text{nonlinear}}(1)}{u_{\text{linear}}(1)}.$$

A value near one is unscreened. A thin shell would give a substantially smaller magnitude.

4 Homogeneous-medium analysis

A local homogeneous potential is useful but, by itself, can be badly misleading for finite objects.

Define

$$t = \frac{\rho p \epsilon}{m_a^2 f_a^2}.$$

For $\epsilon \ll 1$,

$$\frac{V_{\text{eff}}(x) - V_{\text{eff}}(\pi/2)}{V_0} = 1 + \cos 6x - 36t \cos x + O(\epsilon).$$

The stationary condition is

$$\sin 6x = 6t \sin x.$$

4.1 Matter selects an adjacent vacuum globally

The vacuum has six degenerate minima. At any positive visible-matter density,

$$A(\pi/6) < A(\pi/2),$$

so matter makes the adjacent minimum near $x = \pi/6$ globally lower. This does not imply that a finite body can move the field there: it says only which phase an infinite homogeneous medium would prefer.

4.2 Exact local spinodal

The metastable branch connected to $x = \pi/2$ exists while

$$6t \leq \max_{x \in [\pi/3, \pi/2]} \frac{\sin 6x}{\sin x}.$$

Writing $c = \cos x$,

$$\frac{\sin 6x}{\sin x} = 32c^5 - 32c^3 + 6c.$$

The relevant maximum occurs at

$$\cos^2 x_* = \frac{6 - \sqrt{21}}{20},$$

and gives

$$D_* = 1.036340417661276, \quad t_* = \frac{D_*}{6} = 0.172723402943546.$$

For the benchmark,

$$\rho_c \equiv \frac{m_a^2 f_a^2}{p\epsilon} = 267.767 \text{ g cm}^{-3},$$

and hence

$$\rho_{\text{spinodal}} = t_* \rho_c = 46.2497 \text{ g cm}^{-3}.$$

Earth and tungsten stay below this local spinodal. The central solar density in the adopted profile is about 150 g cm^{-3} , so the solar core lies above it. The next section explains why this still does not produce a screened or phase-converted Sun.

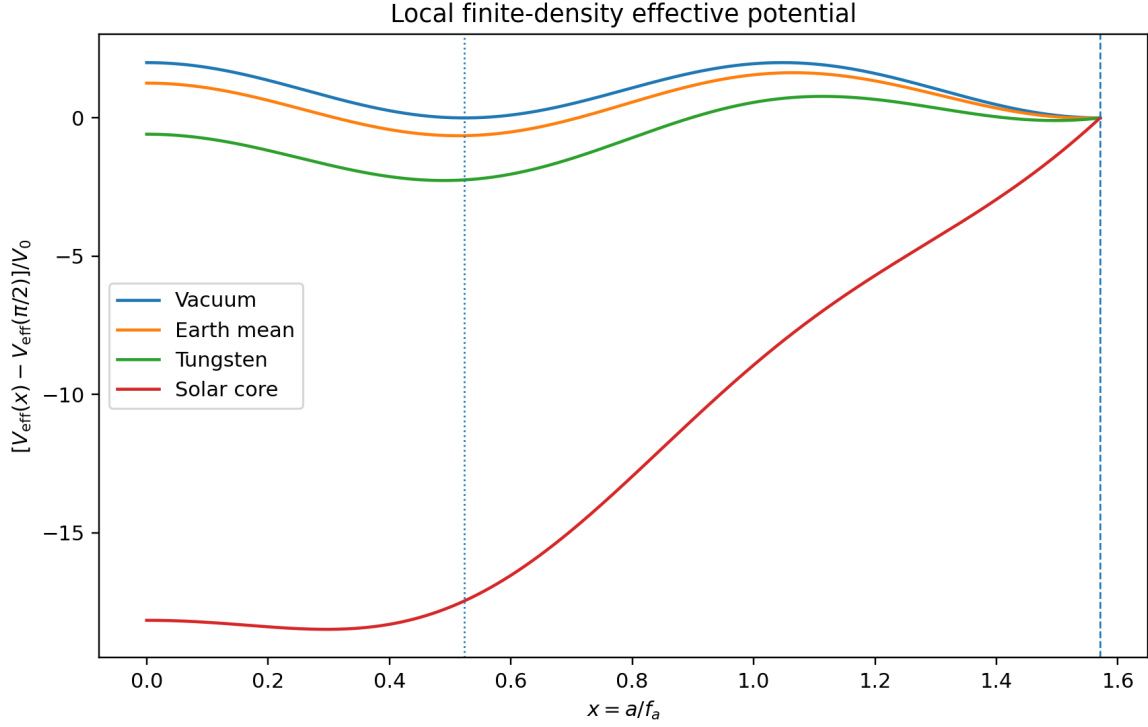


Figure 1: Local homogeneous effective potentials. Visible matter tilts the sixfold vacuum toward the adjacent phase. The solar-core density removes the local branch connected to $x = \pi/2$ in the infinite-medium approximation; this does not imply that a finite Sun can realize that phase.

5 Finite-size phase-conversion bound

Let

$$\Delta x = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$$

be the excursion from the asymptotic vacuum to the adjacent matter-preferred phase.

The maximum rest-energy gain if a spherical source of mass M converts completely is

$$\Delta E_{\text{matter}} = M \Delta A,$$

where

$$\Delta A = 1 - \left(1 - \frac{\sqrt{3}}{2}\epsilon\right)^p.$$

Define

$$C_\epsilon = \frac{\Delta A}{p\epsilon} \longrightarrow \frac{\sqrt{3}}{2} \quad (\epsilon \rightarrow 0).$$

Then

$$\Delta E_{\text{matter}} = 4\pi f_a^2 R q_0 C_\epsilon.$$

5.1 Proposition: exterior-capacity lower bound

If the field reaches the adjacent phase throughout a spherical core of radius R_c and returns to x_∞ outside it, its exterior gradient energy satisfies

$$E_{\text{grad,out}} \geq 2\pi f_a^2 R_c (\Delta x)^2.$$

The minimum is attained, in the massless limit, by the harmonic profile

$$\delta x(r) = -\Delta x \frac{R_c}{r}.$$

The positive vacuum potential and a nonzero scalar mass can only increase the required energy.

For a converted core, define

$$q(R_c) = \frac{M(< R_c) p \epsilon}{4\pi f_a^2 R_c}.$$

A necessary condition for conversion is therefore

$$4\pi f_a^2 R_c q(R_c) C_\epsilon > 2\pi f_a^2 R_c (\Delta x)^2,$$

or

$$q(R_c) > q_{\text{conv}} \equiv \frac{(\Delta x)^2}{2C_\epsilon}.$$

For the benchmark,

$$q_{\text{conv}} = 0.633135.$$

This is the decisive finite-size criterion. The local spinodal depends on density; actual conversion depends on integrated scalar compactness.

5.2 Corollary: no converted phase outside a horizon for the benchmark class

For any concentric region with Newtonian compactness

$$\Phi(R_c) = \frac{GM(< R_c)}{R_c c^2} < \frac{1}{2},$$

one has

$$q(R_c) = 2p\epsilon \left(\frac{M_P}{f_a} \right)^2 \Phi(R_c).$$

Therefore, if

$$f_a \geq M_P, \quad 0 < \epsilon \leq 1, \quad p \leq \frac{2}{27},$$

then

$$q(R_c) < p \leq \frac{2}{27} = 0.0741 < q_{\text{conv}}.$$

Under these assumptions, no nonsingular spherical material region can fully convert to the adjacent phase. The benchmark with $\epsilon = 10^{-6}$ lies vastly deeper in the non-conversion regime.

This theorem does not exclude every possible nonlinear profile, topological defect, cosmological domain history, or highly nonspherical configuration. It does exclude the proposed matter-driven adjacent-vacuum core for ordinary spherical bodies in the stated parameter class.

6 Why there is no thin shell

Chameleon screening requires the field to settle near a density-dependent interior minimum over most of a body, with only a thin transition region contributing to the exterior charge. A standard necessary indicator is

$$m_{\text{eff},\text{in}} R \gg 1.$$

In the present model the relevant curvature scale remains of order m_a throughout the branch explored by Earth, the Sun, and the laboratory source. Their size parameters are

$$\mu_{\oplus} = 1.23 \times 10^{-7}, \quad \mu_{\odot} = 1.34 \times 10^{-5}, \quad \mu_{\text{lab}} = 9.62 \times 10^{-16}.$$

All satisfy

$$\mu \ll 1.$$

The source is therefore much smaller than the field's response length. The field cannot track the local minimum point by point; it responds to the integrated source in the ordinary thick-shell regime.

The finite-density tilt may be large compared with the tiny vacuum barrier in a hypothetical infinite medium, yet the body remains far too small and too weakly compact to pay the spatial gradient cost. This is the key distinction:

local phase preference $\neq$ finite-body field realization.

7 Numerical method

The exact spherical boundary-value problem was solved with an adaptive collocation method.

The scaled variable $u = (x - x_{\infty})/q_0$ avoids loss of precision, since the physical excursions are as small as 10^{-32} for the laboratory benchmark. The exterior domain is represented exactly through the Robin condition at $r = R$.

Three density profiles were used:

1. **Earth:** a smoothed PREM-inspired four-layer profile with inner-core, outer-core, mantle, and crust densities, normalized to a mean density of 5514 kg m^{-3} .
2. **Sun:** an analytic centrally concentrated profile $\rho(r) = \rho_c(1 - r/R)^n$ with $\rho_c = 150 \text{ g cm}^{-3}$ and exponent chosen to reproduce the mean density 1408 kg m^{-3} .

3. **Laboratory source:** a uniform tungsten sphere of radius 5 cm and density 19.25 g cm^{-3} .

The solver independently computes the linear and nonlinear profiles. The nonlinear source terms are retained exactly. Convergence, normalization, asymptotic matching, field-mass conversion, spinodal algebra, and compactness identities are checked in the accompanying verification script.

8 Results

Source	$\mu = m_a R$	q_0	$q_{\text{max}}/q_{\text{conv}}$	δx_c	δx_R	$\mathcal{F}_{\text{scr}}$
Earth	1.226×10^{-7}	1.032×10^{-16}	1.63×10^{-16}	-1.832×10^{-16}	-1.032×10^{-16}	1.000000
Sun	1.339×10^{-5}	3.145×10^{-13}	1.02×10^{-12}	-1.518×10^{-12}	-3.145×10^{-13}	1.000000
W sphere, $R = 5 \text{ cm}$	9.621×10^{-16}	2.218×10^{-32}	3.50×10^{-32}	-3.328×10^{-32}	-2.218×10^{-32}	1.000000

The linear Yukawa surface form factors are

$$\mathcal{F}_Y^\oplus = 0.9999998774,$$

$$\mathcal{F}_Y^\ominus = 0.9999866128,$$

$$\mathcal{F}_Y^{\text{lab}} = 0.99999999999.$$

The nonlinear corrections are bounded by

$$1.70 \times 10^{-22} \quad (\text{Earth}),$$

$$1.41 \times 10^{-18} \quad (\text{Sun}),$$

$$3.08 \times 10^{-38} \quad (\text{laboratory}).$$

Thus the source charges are not thin-shell suppressed.

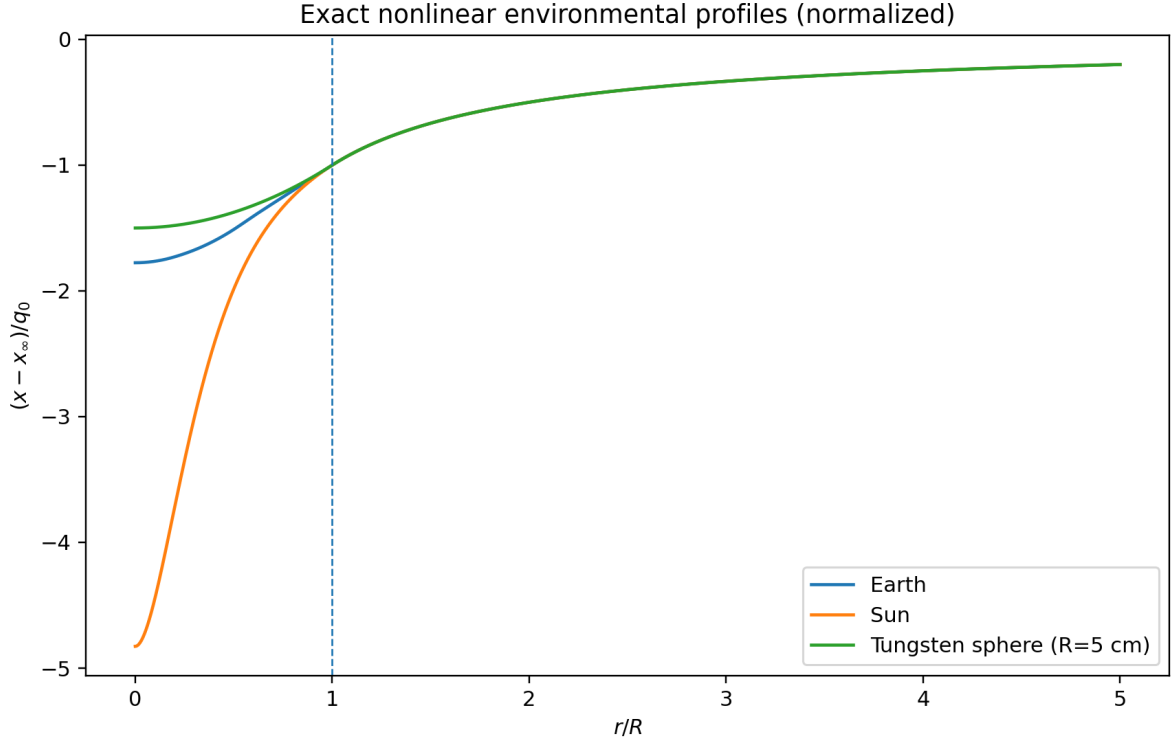


Figure 2: Exact nonlinear environmental profiles normalized by the unscreened scalar compactness q_0 . The dashed line is the source surface. The uniform sphere approaches the analytic massless values $u(0) = -3/2$ and $u(1) = -1$; central concentration deepens the interior solar and terrestrial profiles without suppressing their exterior charge.

9 The solar-core result

The adopted solar profile exceeds the local spinodal density within

$$R_c = 0.1621R_\odot,$$

containing approximately

$$19.64\%$$

of the modeled solar mass.

For that patch,

$$\bar{t}_c = 0.2428 > t_*,$$

so an infinite homogeneous medium at the same mean density would not retain the metastable branch connected to $x = \pi/2$.

However,

$$q_c = 3.81 \times 10^{-13},$$

and therefore

$$\frac{q_c}{q_{\text{conv}}} = 6.02 \times 10^{-13}.$$

Scanning all concentric solar sub-spheres gives the maximum

$$q_{\text{max}} = 6.43 \times 10^{-13}$$

at

$$R_c \simeq 0.368 R_{\odot},$$

still only

$$1.02 \times 10^{-12}$$

of the necessary conversion compactness.

The exact nonlinear solution therefore stays continuously connected to the asymptotic phase. The central excursion is only

$$|x(0) - x_{\infty}| = 1.52 \times 10^{-12}.$$

The apparent solar-core instability was an artefact of applying an infinite-medium effective potential without its gradient terms.

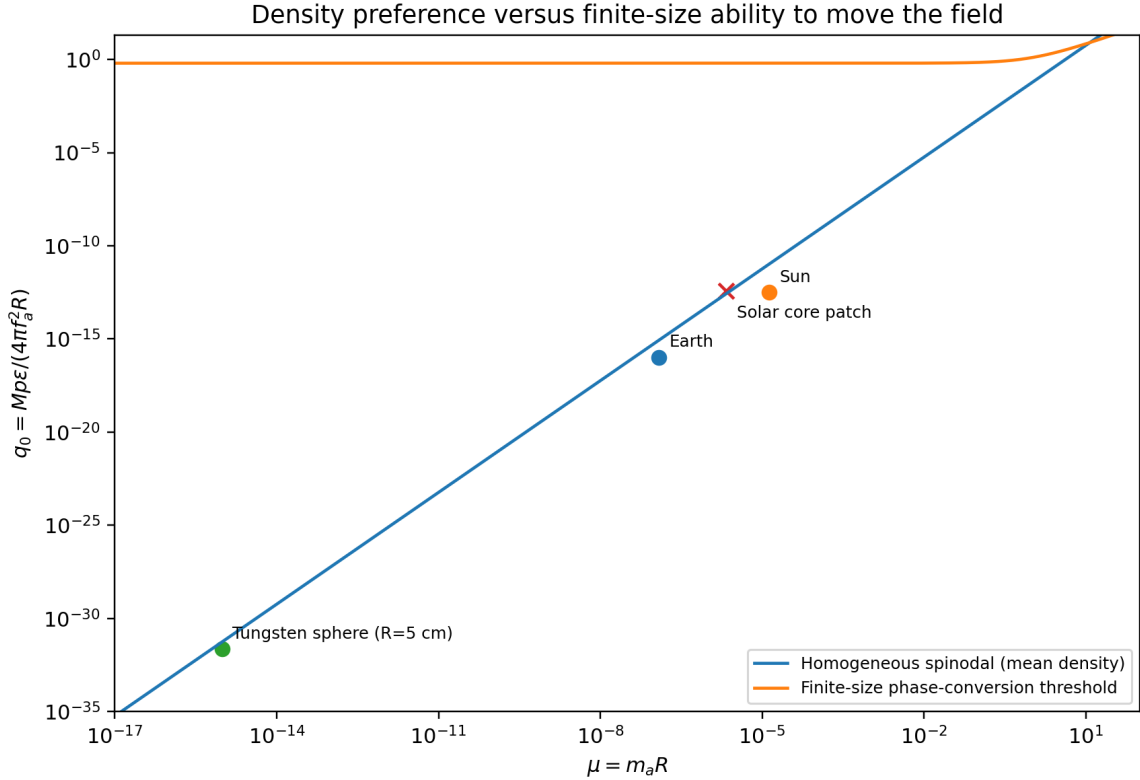


Figure 3: The blue curve marks the homogeneous spinodal inferred from mean density; the orange curve is the finite-size phase-conversion threshold. The solar-core patch lies above the local spinodal but roughly twelve orders of magnitude below the compactness required to move the field into the adjacent phase.

10 Clock and fifth-force signals

At $x_\infty = \pi/2$, the visible QCD coupling is

$$d_g = M_{\text{P}} \frac{\partial}{\partial a} \ln \frac{\Lambda_3}{\chi} = \frac{2}{27} \frac{M_{\text{P}}}{f_a} \epsilon.$$

For the benchmark,

$$d_g = 7.4074 \times 10^{-8}.$$

For a clock-pair sensitivity combination $K_{AB} = \Delta K_\mu - \Delta K_q$, the source-induced ratio shift is

$$\delta \ln \frac{\nu_A}{\nu_B} = K_{AB} d_g \frac{f_a}{M_{\text{P}}} \delta x.$$

For $|K_{AB}| = 1$, the unscreened surface magnitudes are

$$7.64 \times 10^{-24} \quad (\text{Earth}),$$

$$2.33 \times 10^{-20} \quad (\text{Sun}),$$

$$1.64 \times 10^{-39} \quad (5 \text{ cm tungsten sphere}).$$

At Earth's orbit, the solar field gives approximately

$$|\delta x(1 \text{ AU})| = 1.46 \times 10^{-15},$$

and therefore

$$\left| \delta \ln \frac{\nu_A}{\nu_B} \right| \simeq 1.08 \times 10^{-22} |K_{AB}|.$$

A simple eccentric-orbit estimate gives an annual modulation of order

$$3.61 \times 10^{-24} |K_{AB}|.$$

These values are not suppressed by environmental nonlinearities. They are intrinsically small because the protected benchmark uses $\epsilon = 10^{-6}$.

The unscreened universal scalar-force strength before composition factors is approximately

$$\frac{F_a}{F_N} \simeq 2d_g^2 = 1.10 \times 10^{-14}$$

for separations short compared with λ_a . Composition-dependent free-fall signals are further weighted by differences of nuclear dilaton charges.

The v0.5 clock-equivalence-principle consistency relation is therefore retained without an environmental screening factor:

$$\frac{\beta_{AB}}{\eta_{CD}} = -\frac{\Delta K_\mu - \Delta K_q}{\Delta Q'_{\hat{m}}{}^{CD}}$$

within the one-scalar, unscreened, pure-QCD approximation.

11 Robustness and limitations

11.1 Composition

Using $p = 2/27$ assumes the entire source mass scales with Λ_3 . Real atoms contain electron, quark-mass, electromagnetic, and nuclear-binding contributions. Replacing p with species-dependent p_A changes the quantitative charge and equivalence-principle response. Since the QCD fraction is below unity, the benchmark choice generally overestimates the environmental source strength.

11.2 Solar model

The analytic solar profile is not a precision standard-solar-model table. It was chosen to reproduce the correct gross radius, mass, mean density, and a representative central density while retaining a smooth profile. The no-screening result has a margin of twelve orders of magnitude in the conversion criterion, so realistic profile refinements cannot reverse it.

11.3 Pressure and temperature

The environmental source for a relativistic scalar coupling is more precisely the appropriate derivative of the matter free energy and, in many frames, is related to the stress-energy trace. Earth, laboratory matter, and most solar rest mass are nonrelativistic enough that $\rho A(x)$ captures the leading effect. Neutron stars and the early Universe require an equation-of-state treatment.

11.4 Higher-order QCD threshold corrections

The exponent $2/27$ receives calculable perturbative matching corrections. They alter p by relative $O(\alpha_s)$ factors but do not bridge the 10^{12} gap to phase conversion.

11.5 Geometry

The exact BVP calculations are spherical. The capacity argument generalizes qualitatively to arbitrary shapes, but the numerical threshold is shape-dependent through electrostatic capacity. Ordinary laboratory geometries remain fantastically below the required scalar compactness.

11.6 Cosmology and domains

The fact that any positive homogeneous visible density energetically selects an adjacent Z_6 vacuum remains important for cosmological history. A finite star cannot nucleate the phase, but an early homogeneous Universe may move the field when Hubble friction, temperature-dependent sector populations, and causal domain sizes are included. Vacuum/domain selection is now a more serious issue than terrestrial screening.

11.7 Screening mechanisms not included

The calculation excludes additional derivative screening, nonlinear kinetic terms, independent chameleon couplings, plasma-induced terms from hidden-sector populations, and explicit Z_6 -breaking operators. Any ultraviolet completion must be re-evaluated if it introduces such terms.

12 Novelty boundary

Finite-density scalar potentials, density-dependent destabilization, chameleon thin shells, and nonlinear spherical profile methods are established subjects.

The candidate contribution here is narrower:

1. the exact environmental equation for the protected visible 2/27 QCD threshold model;
2. the separation between the homogeneous spinodal and finite-size scalar compactness;
3. the capacity lower bound applied to adjacent Z_6 phase conversion;
4. the result that the solar core can lie above the local spinodal while the full nonlinear Sun remains unscreened and unconverted by twelve orders of magnitude;
5. preservation of the chronometric clock–equivalence-principle consistency line without a screening factor.

A complete priority claim would still require backward and forward citation analysis across finite-density axion, symmetron, chameleon, scalarization, and false-vacuum bubble literature.

13 Acceptance matrix

Requirement	Verdict	Evidence
Exact nonlinear environmental equation	PASS	Derived from $V_{Z_6} + \rho A(x)$
Earth boundary-value solution	PASS	Adaptive nonlinear BVP converged
Solar boundary-value solution	PASS	Adaptive nonlinear BVP converged
Laboratory-source solution	PASS	Adaptive nonlinear BVP converged
Thin-shell screening	ABSENT	$m_{\text{eff}} R \ll 1$, $\mathcal{F}_{\text{screen}} = 1$
Adjacent-phase conversion	EXCLUDED for benchmark spherical sources	Capacity bound and $q_{\text{max}}/q_{\text{conv}} \ll 1$
Solar-core local destabilization	PRESENT in homogeneous approximation	$\rho_{\text{core}} > \rho_{\text{spinodal}}$
Solar-core actual conversion	ABSENT	$q_c/q_{\text{conv}} = 6.02 \times 10^{-13}$
Visible 2/27 response	PRESERVED	No environmental charge suppression
Observable benchmark clock signal	NO	Unscreened but $\lesssim 10^{-20}$ at solar surface
Cosmological vacuum selection	OPEN	Requires finite-temperature FRW evolution
Nonspherical and multi-body validation	OPEN but unlikely to alter benchmark verdict	Huge compactness margin

Requirement	Verdict	Evidence
Neutron-star treatment	OPEN	Requires relativistic EOS and full GR

14 Next decisive step

Environmental screening is not the bottleneck for the protected benchmark. The next calculation should therefore combine two tasks:

1. **Cosmological vacuum selection:** solve

$$\ddot{x} + 3H\dot{x} - \frac{\nabla^2 x}{a_{\text{FRW}}^2} = -\frac{1}{f_a^2} \frac{\partial V_{\text{eff}}(x, T, \rho_k)}{\partial x}$$

through reheating, replicated-sector decoupling, the QCD crossover, and matter domination to determine whether the Universe lands in the visible $x_\infty = \pi/2$ branch without unacceptable walls or isocurvature.

2. **Observable-coupling optimization:** vary (M, f_a, ϵ, N) subject to radiative protection, fifth-force limits, clock limits, cosmology, and the no-conversion condition to determine the largest allowed nonzero chronometric shear.

The environmental result is conceptually favourable:

the source remains visible to the field; it is the coupling, not screening, that limits the signal.

15 References

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